

Lesson 28 — Evaluating & Improving a Model (Data Literacy)

Matches LEGO Teacher Portal lesson: "Skateboard Stunts".

Objective (I can...): I can test my model's accuracy, find where it fails, and improve it by changing the dataset. **CS concept:** Model evaluation; refining a model with better/more balanced data · **Builds on:** Lesson 27 (using a model). *Privacy: on-device in the browser, no login, images not saved.*

AI Connection (2 min): This is why "more and better examples help": a model is only as good as its dataset. *First-principles question:* "If it keeps confusing two categories, is the fix more code — or better data? How would you tell?"

Materials: Device per pair (Teachable Machine), LEGO CS & AI kit, varied test items, notebooks (dataset log continues). **Model reuse:** AI responder model from Lesson 26 — retrain and code.

60-minute flow - Warm-up (8): "Fooled you" — coach shows a classifier (or describes one) that fails on an edge case; pairs hypothesize what was missing from its training data. - **Concept + demo (10):** Demo a weak classifier, run a quick accuracy check on held-back test items, identify the failing category, add better/more varied examples, retrain, and show accuracy improve. Introduce *balanced dataset*. - **Build/Code (30):** Pairs run their own accuracy check (e.g., 10 test trials per category, tally correct in the dataset log), diagnose the weakest category, add targeted examples (varied angle, lighting, background), retrain, and re-test. They record before/after accuracy. *Swap at 15 min.* - **Challenge (7):** Introduce a tricky look-alike category and improve the dataset until the model handles it. - **Share + discussion (5):** Pairs report accuracy change, then discuss **why** the model still errs — was the dataset unbalanced, trained in one lighting, or on only one person? Name one better example that would help. Model stays built.

Pair roles: Driver captures new examples, Navigator runs the accuracy trials and records the tally. Swap at 15 min. (*Trio: third owns the test protocol.*)

Differentiation: *Support* — improve one category and measure just that. *Extension / fast-finisher* — balance three categories and report a per-category accuracy. *Reteach* — add examples one at a time, re-testing so each improvement is visible.

Exit check: Pair shows measured improvement after changing the dataset and can name one data-quality reason the model erred. **At-home (optional):** Explain to a family member, with your own example, why an AI trained on a narrow dataset can be unreliable. ~10 min.